COMP4121 Advanced Algorithms

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Some Introductory Notes

A few basic fact from Statistics

- **A random vector** is a sequence of n random variables $(X_1, \ldots, X_n)$.

- We denote the (multivariate) PDF of such a vector by $f_{\mathbf{X}}(x_1, \ldots, x_n)$.

- Random variables $X_1, \ldots, X_n$ are **independent** if for all subsets $A_1, \ldots, A_i \subset \mathbb{R}$,

$$\mathcal{P}(X_1 \in A_1, \ldots, X_n \in A_n) = \prod_{i=1}^{n} \mathcal{P}(X_i \in A_i). \tag{1}$$

- This happens just in case

$$f_{\mathbf{X}}(x_1, \ldots, x_n) = \prod_{i=1}^{n} f_{X_i}(x_i). \tag{2}$$

A few basic fact from Statistics

- If random variables $X_1, \ldots, X_n$ are independent and have the same probability distribution $F(x)$, we say that $X_1, \ldots, X_n$ are **IID (independent identically distributed)** random variables, or **random sample of size n from F**.

- The **sample mean** is the random variable defined by

$$\overline{X} = \frac{X_1 + \ldots + X_n}{n}$$

- Assume $X_1, \ldots, X_n$ are IID with mean $E(X_i) = \mu$ and variance $V(X_i) = \sigma^2$; then, since X_i are equally distributed, the expected value of $\overline{X}$ and the variance $V(\overline{X})$ satisfy

$$E(\overline{X}) = \frac{1}{n} \sum_{i=1}^{n} E(X_i) = \frac{1}{n} n\mu = \mu \tag{3}$$

A few basic fact from Statistics

- Using this we get

$$V(\overline{X}) = E(\overline{X} - E(\overline{X}))^2 = E(\overline{X} - \mu)^2 = E\left(\frac{\sum_{i=1}^{n}(X_i - \mu)}{n}\right)^2 =$$

$$E\left(\frac{1}{n^2}\sum_{i=1}^{n}(X_i - \mu)^2 + \frac{2}{n^2}\sum_{i \neq j}^{n}(X_i - \mu)(X_j - \mu)\right) =$$

$$\frac{1}{n^2}\left(\sum_{i=1}^{n}E(X_i - \mu)^2 + 2\sum_{i \neq j}^{n}E((X_i - \mu)(X_j - \mu))\right);$$

- We now use the fact that if X_i, X_j are independent then their covariance is 0:

$$E((X_i - \mu)(X_j - \mu)) = E(X_i - \mu)E(X_j - \mu) = 0 \times 0 = 0$$

- Thus

$$V(\overline{X}) = \frac{1}{n^2}\sum_{i=1}^{n} E(X_i - \mu)^2 = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}. \tag{4}$$

- Note that this explains why taking the mean of multiple measurements provides more accurate value than a single measurement: if the variance of a single measurement is σ^2 then the variance of the mean of n measurements is only σ^2/n.

A few basic fact from Statistics

- Assume now that we have n measurements X_i of a quantity using the same instrument and want to estimate the variance of the instrument.

- Since the expected value of $\overline{X}$ is μ one might think that

$$E\left(\frac{1}{n}\sum_{i=1}^{n}(X_i-\overline{X})^2\right)\stackrel{??}{=}E\left(\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu)^2\right)=\frac{1}{n}\sum_{i=1}^{n}E(X_i-\mu)^2=\frac{n\,\sigma^2}{n}=\sigma^2$$

- This would make $\frac{1}{n}\sum_{i=1}^{n}(X_i-\overline{X})^2$ an unbiased estimator for the variance of each of X_i.
- But this is not quite so - the "equality" with the question marks fails, because we do not take into account that in fact $\overline{X}$ is not quite equal to μ, making the expected value of the lefthand side slightly smaller than σ^2 because $\eta=\overline{X}$ in fact minimises the value of $s(\eta)=\sum_{i=1}^{n}(X_i-\eta)^2$.

A few basic fact from Statistics

- An unbiased estimate of the variance of all X_i is given by the **sample variance** S_n^2 defined as

$$S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2. \tag{5}$$

To see that the expected value of the sample variance is equal to the variance σ^2 of all X_i, note that

$$E\left(\sum_{i=1}^{n}(X_i - \overline{X})^2\right) = E\left(\sum_{i=1}^{n}(X_i^2 - 2X_i\overline{X} + \overline{X}^2)\right) = \tag{6}$$

$$E\left(\sum_{i=1}^{n}X_i^2 - 2\sum_{i=1}^{n}X_i\overline{X} + \sum_{i=1}^{n}\overline{X}^2\right) = E\left(\sum_{i=1}^{n}X_i^2 - 2n\overline{X}^2 + n\overline{X}^2\right) =$$

$$E\left(\sum_{i=1}^{n}X_i^2 - n\overline{X}^2\right) = \sum_{i=1}^{n}E(X_i^2) - n\,E(\overline{X}^2) = nE(X_1^2) - n\,E(\overline{X}^2) \tag{7}$$

A few basic fact from Statistics

- We now use the fact that for every random variable Y with mean μ and standard deviation σ we have

$$\begin{aligned} \sigma^2 = & E(Y-\mu)^2 = E(Y^2 - 2\mu Y + \mu^2) = E(Y^2) - 2\mu E(Y) + \mu^2 = \\ & E(Y^2) - 2\mu^2 + \mu^2 = E(Y^2) - \mu^2 \end{aligned} \tag{8}$$

Thus, $E(Y^2) = \sigma^2 + \mu^2$. Continuing (**??**) and using (**??**) and (**??**)

$$\begin{aligned} E\left(\sum_{i=1}^{n}(X_i - \overline{X})^2\right) & = nE(X_1^2) - n\,E(\overline{X}^{\,2}) = n(\sigma^2+\mu^2) - n\,E(\overline{X}^2) \\ & = n(\sigma^2+\mu^2) - n\left(E(\overline{X}-\mu)^2 + \mu^2\right) = \\ & = n(\sigma^2+\mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) = \\ & = (n-1)\sigma^2 \end{aligned} \tag{9}$$

- This clearly implies that $E(S_n^2) = E\left(\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2\right) = \sigma^2$, i.e., that S_n^2 is an unbiased estimator for the variance of X.

A tool we will often need: Markov Inequality

- Assume $X > 0$ is a non-negative random variable;
- assume also that $t > 0$ is any positive real number;

  then:
  $$P\{X \geq t\} \leq \frac{E[X]}{t}$$
- **Proof:** Essentially, we take into account only events when $X \geq t$ and ignore events when $X < t$:
- If X is discrete, then

$$\begin{aligned}
E[X] &= \sum_{v} P\{X = v\} \cdot v = \sum_{0 \leq v < t} P\{X = v\} \cdot v + \sum_{v \geq t} P\{X = v\} \cdot v \\
&\geq \sum_{v \geq t} P\{X = v\} \cdot v \geq \sum_{v \geq t} P\{X = v\} \cdot t = t \sum_{v \geq t} P\{X = v\} \\
&= t\, P\{X \geq t\}
\end{aligned}$$

- Divide now both sides by $t > 0$ to obtain the Markov Inequality.
- The case when X is continuous is essentially identical and you can find it in the probability refresher lecture notes.

Application of Markov's Inequality: Chebyshev's Inequality (Markov was a student of Chebyshev)

- **Theorem:** Let X be any random variable with a finite expectation $E[X]$ and a finite variance $V[X]$. Then

$$\mathcal{P}(|X - E[X]| \geq t) \leq \frac{V[X]}{t^2}$$

- **Proof:** Note that

$$\mathcal{P}(|X - E[X]| \geq t) = \mathcal{P}((X - E[X])^2 \geq t^2)$$

- Also, $Y = (X - E[X])^2$ is a non-negative random variable with a finite expectation $E[Y] = E((X - E[X])^2) = V[X]$.
- Thus, we can apply the Markov inequality to Y to obtain

$$\mathcal{P}(|X - E[X]| \geq t) = \mathcal{P}((X - E[X])^2 \geq t^2) = \mathcal{P}(Y > t^2) \leq \frac{E[Y]}{t^2} = \frac{V[X]}{t^2}$$

Law of Large Numbers

- If we get a large number of independent samples $X_1, \ldots, X_n$ of a random variable X with an expectation $E[X]$ we would expect that the mean of these samples should be close to the expected value $E[X]$ of X. The Law of large numbers bounds the probability that such a mean is further away from $E[X]$ than a prescribed value ε:

$$\mathcal{P}\left(\left|\frac{X_1 + X_2 + \ldots + X_n}{n} - E[X]\right| > \varepsilon\right) \leq \frac{V[X]}{n\varepsilon^2}$$

- **Proof:** By the Chebyshev inequality,

$$\mathcal{P}\left(\left|\frac{X_1 + \ldots + X_n}{n} - E\left[\frac{X_1 + X_2 + \ldots + X_n]}{n}\right]\right| > \varepsilon\right) \leq \frac{V\left[\frac{X_1+X_2+\ldots+X_n]}{n}\right]}{\varepsilon^2}$$

- We now use the fact that the expectation is a linear operator and all X_i are equally distributed to conclude that

$$E\left[\frac{X_1 + X_2 + \ldots + X_n}{n}\right] = \frac{E\left[X_1\right] + E\left[X_2\right] + \ldots + E\left[X_n\right]}{n} = \frac{n\,E[X]}{n} = E[X]$$

- On the other hand, variance of any random variable X satisfies $V[a\,X] = a^2 V[X]$.
- Also, for INDEPENDENT variables $X_1, \ldots, X_n$ we have $V[X_1 + \ldots + X_2] = V[X_1] + \ldots + V[X_n]$.

Law of Large Numbers

- Thus

$$\mathcal{P}\left(\left|\frac{X_1+\ldots+X_n}{n} - E\left[\frac{X_1+\ldots+X_n]}{n}\right]\right| > \varepsilon\right) \leq \frac{V\left[\frac{X_1+\ldots+X_n]}{n}\right]}{\varepsilon^2}$$

implies

$$\mathcal{P}\left(\left|\frac{X_1+\ldots+X_n}{n} - E[X]\right| > \varepsilon\right) \leq \frac{\frac{n\,V[X]}{n^2}}{\varepsilon^2} = \frac{V[X]}{n\,\varepsilon^2}$$

- To summarise, if $X_1, \ldots, X_n$ are independent, equally distributed random variables with a finite expectation $E[X_i] = \mu$ and variance $V[X_i] = v$, then

$$E\left[\frac{X_1+\ldots+X_n}{n}\right] = \mu;$$

$$V\left[\frac{X_1+\ldots+X_n}{n}\right] = \frac{v}{n}$$

$$\mathcal{P}\left(\left|\frac{X_1+\ldots+X_n}{n} - \mu\right| > \varepsilon\right) \leq \frac{v}{n\epsilon^2}$$

- Let A be an $n \times n$ matrix. We say that λ is an *eigenvalue* and that a vector $\vec{x}$ is an *eigenvector* if

$$A\vec{x} = \lambda\vec{x}$$

- This happens if and only if

$$(A - \lambda I)\vec{x} = 0$$

  where I is an $n \times n$ identity matrix.
- This happens just if

$$\det(A - \lambda I) = 0$$

Basic Linear Algebra

Note that $\det(A - \lambda I) =$

$$\det \begin{pmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{pmatrix} = \begin{vmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{vmatrix}$$

- Thus, $\det(A - \lambda I) = 0$ is of the form $P(\lambda) = 0$ where $P(\lambda)$ is a polynomial in λ of degree n.
- Every matrix of size $n \times n$ has n eigenvalues which are generally complex numbers.
- However, if a matrix with real coefficients is symmetric, then its eigenvalues are real numbers.

- A matrix A is positive-semidefinite if for every vector $\vec{x}$ we have

$$\vec{x}^\top A\vec{x} \geq 0$$

- If a symmetric real matrix is also positive-semidefinite then all of its eigenvalues are non negative reals.